Solution :

The average number of typographical errors per page in the book is given by

$\lambda =$ (390/520) = 0.75.

Hence using Poisson probability law, the probability of x errors per page is given by

$$P(X=x) = \frac{e^{-\lambda}\lambda^{x}}{x!} = e^{-0.75}\frac{(0.75)^{x}}{x!}, x = 0,1,2,\ldots$$

The required probability that a random sample of 5 pages will contain no error is given by :

$$[P(X=0)]^{5} = \left(e^{-0.75}\right)^{5} = e^{-3.75}$$

Example 7.16

An insurance company has discovered that only about 0.1 per cent of the population is involved in a certain type of accident each year. If its 10,000 policy holders were randomly selected from the population, what is the probability that not more than 5 of its clients are involved in such an accident next year? ($e^{-10} = .000045$)

Solution :

p = probability that a person will involve in an accident in a year

= 0.1/100 = 1/1000

given n = 10,000

so, $\lambda = np = 10000\left(\frac{1}{1000}\right) = 10$

Probability that not more than 5 will involve in such an accident in a year

$P(X \leq 5) = P(X=0)+P(X=1)+P(X=2)+P(X=3)+P(X=4)+P(X=5)$

$$= e^{-10}\left[1+\frac{10}{1!}+\frac{10^{2}}{2!}+\frac{10^{3}}{3!}+\frac{10^{4}}{4!}+\frac{10^{5}}{5!}\right]$$

= 0.06651

Example 7.17

One fifth percent of the the blades produced by a blade manufacturing factory turn out to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing no defective, one defective and two defective blades respectively in a consignment of 1,00,000 packets ($e^{-0.02}$ =.9802).

Solution :

$P = 1/5/100 = 1/500 = 0.002 \quad n = 10 \;.\; \lambda = np = 0.02$

$$p(x) = \frac{e^{-\lambda}\lambda^{x}}{x!} = \frac{e^{-0.02}(0.02)^{x}}{x!}$$

(i) Number of packets containing no defective

$= N \times p(\text{o}) = 1,00,000 \times e^{-0.02}$

= 98020

(ii) Number of packets containing one defective

$= N \times p(1) = 1,00,000 \times 0.9802 \times 0.02$

= 1960

(iii) Number of packets containing 2 defectives

$= N \times p(2) = 20$

Example 7.18

If the probability that an individual suffers a bad reaction from injection of a given serum is 0.001, determines the probability that out of 2,000 individuals (a) exactly 3 and (b) more than 2 individuals will suffer a bad reaction.

Solution :

Consider a 2,000 individuals getting injection of a given serum , n = 2000

Let X be the number of individuals suffering a bad reaction.

Let p be the probability that an individual suffers a bad reaction = 0.001

and $q = 1 - p = 1 - 0.001 = 0.999$

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